Some Applications of Natural Language Processing

1. Chatbots

Chatbots are a form of artificial intelligence that are programmed to interact with humans in such a way that they sound like humans themselves. Depending on the complexity of the chatbots, they can either just respond to specific keywords or they can even hold full conversations that make it tough to distinguish them from humans. **Chatbots are created using Natural Language Processing and Machine Learning,** which means that they understand the complexities of the English language and find the actual meaning of the sentence and they also learn from their conversations with humans and become better with time. **Chatbots work in two simple steps. First, they identify the meaning of the question asked and collect all the data from the user that may be required to answer the question. Then they answer the question appropriately.**

2. Autocomplete in Search Engines

Have you noticed that **search engines tend to guess what you are typing and automatically complete your sentences?** For example, On typing "game" in Google, you may get further suggestions for "game of thrones", "game of life" or if you are interested in maths then "game theory". All these suggestions are provided using autocomplete that uses Natural Language Processing to guess what you want to ask. Search engines use their enormous data sets to analyze what their customers are probably typing when they enter particular words and suggest the most common possibilities. **They use Natural Language Processing to make sense of these words and how they are interconnected to form different sentences.**

3. Voice Assistants

These days voice assistants are all the rage! Whether its **Siri, Alexa, or Google Assistant,** almost everyone uses one of these to **make calls, place reminders, schedule meetings, set alarms, surf the internet, etc**. These voice assistants have made life much easier. But how do they work? They use a complex combination of speech recognition, natural language understanding, and natural language processing to understand what humans are saying and then act on it. The long term goal of voice assistants is to become a bridge between humans and the internet and provide all manner of services based on just voice interaction. However, they are still a little far from that goal seeing as Siri still can't understand what you are saying sometimes!

4. Language Translator

Want to translate a text from English to Hindi but don't know Hindi? Well, Google Translate is the tool for you! While it's not exactly 100% accurate, it is still a great tool to convert text from one language to another. **Google Translate and other translation tools as well as use Sequence to sequence modeling that is a technique in Natural Language Processing.** It allows the algorithm to convert a sequence of words from one language to another which is translation. Earlier, language translators used Statistical machine translation (SMT) which meant they analyzed millions of documents that were already translated from one language to another (English to Hindi in this case) and then looked for the common patterns and basic vocabulary of the language. However, this method was not that accurate as compared to Sequence to sequence modeling.

5. Sentiment Analysis

Almost all the world is on social media these days! And companies can use sentiment analysis to understand how a particular type of user feels about a particular topic, product, etc. They can use natural language processing, computational linguistics, text analysis, etc. to understand the general sentiment of the users for their products and services and find out if the sentiment is good, bad, or neutral. Companies can use sentiment analysis in a lot of ways such as to find out the emotions of their target audience, to understand product reviews, to gauge their brand sentiment, etc. And not just **private companies, even governments use sentiment analysis to find popular opinion and also catch out any threats to the security of the nation.**

6. Grammar Checkers

Grammar and spelling is a very important factor while writing professional reports for your superiors even assignments for your lecturers. After all, having major errors may get you fired or failed! That's why grammar and spell checkers are a very important tool for any professional writer. They can not only correct grammar and check spellings but also suggest better synonyms and improve the overall readability of your content. And guess what, they utilize natural language processing to provide the best possible piece of writing! The NLP algorithm is trained on millions of sentences to understand the correct format. That is why it can suggest the correct verb tense, a better synonym, or a clearer sentence structure than what you have written. Some of the most **popular grammar checkers that use NLP include Grammarly, WhiteSmoke, ProWritingAid, etc.**

7. Email Classification and Filtering

Emails are still the most important method for professional communication. However, all of us still get thousands of promotional Emails that we don't want to read. Thankfully, our emails are automatically divided into 3 sections namely, Primary, Social, and Promotions which means we never have to open the Promotional section! But how does this work? **Email services use natural language processing to identify the contents of each Email with text classification so that it can be put in the correct section.** This method is not perfect since there are still some Promotional newsletters in Primary, but its better than nothing. In more advanced cases, some companies also use specialty anti-virus software with natural language processing to scan the Emails and see if there are any patterns and phrases that may indicate a phishing attempt on the employees.

The Timeline of NLP Evolution

• **1950s**: The first traces of NLP came in the 1950s, when rule-based methods were used to build NLP systems. These were primarily focused on Machine Translation, and came into high demand as a result of World War II and the need for effective translations. Uses included word/sentence analysis, question-answering, and machine translation.

• **1980s**: Computational grammar became an active field of research. Grammar tools and resources became more available and in demand.

• **1990s**: The 1990s saw a booming development of the web and a new interest in artificial intelligence. This created an abundance of knowledge and drove statistical learning methods to work on NLP tasks. Statistical learning learns from a specific dataset and describes its features.

• **2012**: Deep Learning took over statistical learning, producing drastic improvements in the NLP system. Deep Learning deep dives into raw data and learns its attributes.

• **Current day**: There's a huge demand for machines that can talk and understand our needs, and NLP is the key to that door. Just look at products like Alexa and chatbots. The neural network-based NLP (referred to as 'neural NLP') framework has achieved new levels of quality and has become the governing approach for NLP. Deep learning makes tasks such as MT, machine reading comprehension (MRC), chatbot, etc. considerably easier. There are so many important use cases of NLP in the world now.

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